

SUSPENDED

GA in BSc (Hons) Data Science with Artificial Intelligence
School of Computing, Engineering and Technology

Version	2.0
Course Type	Undergraduate
Award Type	Graduate Apprenticeship in Bachelor of Science with Honours - GA in BSc (Hons)
Awards	
No Course Award Assigned	
Code	0718
SCQF Level	Level 10
QAA Subject Benchmark	Computing (2022)
SCQF Credit Points	480
ECTS Credit Points	240
Valid From	2026 September

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100755	data management	data management	50%
100359	artificial intelligence	artificial intelligence	50%
Total			100%

Professional, Statutory and Regulatory Body (PSRB)

No PSRBs have been recorded

The four-year Graduate Apprenticeship in BSc (Hons) Data Science with Artificial Intelligence is an industry focused degree route, where students work in course-relevant employment for a partner employer while studying towards a fully accredited degree. The course combines workplace training with a programme of education designed to meet students' career ambitions. The core curriculum is focused on latest thinking in data science coupled with a "flipped-classroom" approach to work-based learning developed through ongoing collaboration between University staff, Workplace Mentors, and line managers in the workplace.

Learning support materials are provided through the VLE which includes a range of equipment such as video and online learning technologies as well as online interaction. Where appropriate, modules use technologies such as computer-mediated formative assessment to bring interactivity and immediacy to the teaching and learning process. Additionally, there

What the Course Involves	are opportunities for the student to attend scheduled on-campus events to allow face-to-face interaction with staff and peers, participate in group activities with peers and this contributes to the overall student experience.	
	Staff support students engaged in problem solving from an academic and/or theoretical perspective, while Workplace Mentors guide students in the application of theory to workplace scenarios. University staff undertake visits to the student's place of work in order to observe the work-based learning environment and discuss the student's progress with the Workplace Mentor.	
	This combination of blended academic study allows the student to contextualise learning immediately within their workplace thereby developing their understanding and enhancing and their contribution to organisational effectiveness.	
	Each stage has four 30-credit modules, with two modules delivered concurrently in Semester 1, and two modules delivered concurrently in Semester 2, both in 18-week learning blocks. The exception to this is the Stage 4 Honours Individual module CM4700 which runs 'long and thin' throughout the two semesters of Stage 4 - this shown in the structure below.	
	Module Delivery Structure	
	Stage	1
	Semester 1	CM1705 Fundamentals of Programming CM1706 Introduction to Business Analytics
	Semester 2	CM1704 Analytical and Problem Solving Skills CM1707 Professional Development
	Stage	2
	Semester 1	CM2710 Advanced Programming CM2709 Databases and Information Security
	Semester 2	CM2712 Data Mining CM2705 Business Ethics and Management
	Stage	3
	Semester 1	CM3710 Machine Learning and Artificial Intelligence

		CM3708 Big Data Systems
	Semester 2	CM3709 Business Intelligence CM3704 Business Ideas and Opportunities
	Stage	4
	Semester 1	CM4706 Advanced Data Science CM4708 Data Science Trends and Applications
	Semester 2	CM4709 Computer Vision
	YEAR (long and thin)	CM4700 Honours Individual Project
UCAS Code	Code	
Course Collaborations		
No collaborations listed		
	Graduate Apprenticeships provide work-based learning opportunities for employees and have been created by Skills Development Scotland in partnership with industry and the Further and Higher education sector. Graduate Apprenticeships combine academic knowledge with work-based learning to enable students to become more effective and productive in the workplace. They provide a new way into degree-level study for individuals who are currently employed, and afford the opportunity to undertake degree study whilst working and earning for those individuals who may otherwise have sacrificed Secondary or Further education to enter the world of work. They enable employers to develop their workforce and support staff to build their skills to industry and professional standards. Since Graduate Apprenticeships are designed around the needs of industry, employers can have confidence that what staff are learning will directly contribute to the success of the business. As they are in continuing employment, students can directly apply academic learning to workplace situations, thus truly bringing their studies to life.	
	The Graduate Apprenticeship in BSc (Hons) Data Science with Artificial Intelligence provides students with the knowledge, understanding and skills required to become a successful professional data scientist. The course aims to develop the student's intellectual capabilities, professional understanding and judgement, and personal, interpersonal, project management, problem solving and communications skills.	
	The course covers topics which offer students a thorough grounding in data analytics, data management, data engineering, machine learning and artificial intelligence. Students will be	

Introduction to the Course	able to help their organisations gain insight from their data to drive innovation and efficiency. They will have the requisite underpinning technical knowledge and will develop the lifelong learning professional skills, ethics and behaviours required to stay abreast of rapidly evolving technologies in data science and machine learning. Depending on the nature of the organisation, typical examples of career opportunities across a broad range of sectors, might include Data Scientist; Data Analyst; Data Engineer; Business Intelligence Professional; Business Analytics Specialist; Database Administrator, AI Engineer; or AI Solutions Architect. However, in many organisations, there may be specific elements of the degree content involved in the role which are not reflected in the job title but which are a key component within it. Examples include Marketing Analysts and Customer Relations Managers. The Graduate Apprenticeship allows the student to balance academic and workplace learning effectively. With the University's supported online learning platform and tools, students have flexibility to engage in their academic studies around personal and organisational commitments. Attendance on campus is minimised to reduce workplace disruption and costs. Online learning can be accessed 24/7, is varied and relevant to the topics undertaken and includes the use of multi-media interactive activities, podcasts, quizzes, webinars and forum discussions, thus providing an engaging and motivating learning experience. On-campus workshops will introduce and consolidate online learning, and support the student in preparation for academic assessment. Students are supported through the course to achieve professional excellence through a suite of modules designed to develop academic, personal and interpersonal skills. They will be encouraged to develop an enquiring, analytical, open minded and creative approach to foster independent judgement, critical self-awareness, ability to reflect, and an appreciation of the wider context of the course of study. These skills will be developed and practiced both in the context of the academic environment and in their workplace organisation.
Educational Aims of the Course	The purpose of the Graduate Apprenticeship in BSc (Hons) Data Science with Artificial Intelligence is to provide students with the required knowledge and understanding of data science and artificial intelligence (AI) principles and techniques. The development of this knowledge is reinforced with the development of strong personal, interpersonal and problem solving team, enabling students to become professional data analysts equipped to perform effectively in any working environment. Specifically, the aims of the course are as follows: - To equip students with a range of analytical and problem solving skills used in data science and AI. - To provide students with required understanding of data science and AI principles to cope adequately with technological change. - To develop the students' ability to contribute to the specification, design, testing, commissioning and maintenance of data engineering systems both generally and within the context of the work-based environment. - To develop the students' ability in the use of data science and AI techniques and tools. - To make the students aware of the ethical, social and economic impact of data science and AI. - To develop the students' interpersonal skills to enable effective communication in the working environment. - To integrate the programme with the students' experiential learning and training as part of an apprenticeship with their employers. - To develop the students' commitment to CPD and reflective practice such that they constantly seek to learn from the connection of theory to practice. Graduates of the course will possess the following attributes: - An understanding of the range of techniques and tools available for the development of data science and AI solutions. - The ability to source, collect, clean and integrate large data sets of various types and sources. - The ability to analyse and visualise complex data sets using advanced statistical and data mining techniques. - A strong understanding of machine learning and artificial intelligence, enabling students to design intelligent systems to respond to technological development and innovations.

	- The ability to utilise modern data science software applications and programming to develop solutions for business analytics. - An awareness of the ethical, social, regulatory and economic impact of data science and AI solutions. - The ability to effectively deploy technical, project management and business skills during the course and apply them to a work-based data science project. - The ability to integrate theory with professional practice via experiences gained within the work-based learning environment, critically evaluating solutions based on experiences within that environment. The Graduate Apprenticeship student journey differentiates itself from the traditional course of study in that students will be employed for the entire four-year duration of their educational journey. Their employment environment affords them a unique opportunity to take full advantage of work-based learning opportunities, appreciate the relevance of theoretical learning, and open up future career opportunities with their employer. Activities within the employing organisation are coordinated to reflect the content of the modules in which students are enrolled, and are designed to provide the practical application of the theory received through academic learning. The work-based learning within the course aims to offer a new learning experience to meet the needs of both individuals and employers. This will play a role in achieving the aspiration of the Scottish Government to see a culture of real partnership between employers and education and increasing skill levels in the workforce. This is entirely in keeping with the commitments of the University to 'extend the reach and relevance of learners' opportunities to gain employment and thrive in their professional career'; to 'develop the range and diversity of the University's provision to support individuals' professional development'; to 'ensure a high quality student experience'; to 'adopt a student-centred approach to curriculum design and delivery'; to 'work with organisations to enhance skills'; and to 'expand the range and depth of partnerships with employers and the professions'.
Awarding Institution	Robert Gordon University
Course Duration	Work-Based Learning Full-Time - 4 years with a September intake. 5 years with a February intake (availability subject to funding as per Course Specific Academic Regulations).
Distinctive Features of the Course	The course integrates the expertise of staff gained from research, consultancy and scholarly activity into the programme materials. The School has an excellent track record in data science and AI projects conducted in collaboration with industry, and those will offer a wealth of case studies that will enrich the students' learning experience and provide real-world scenarios of applications of data science in the workplace. Delivery of the course is further enhanced by the opportunity to hear from guest speakers with illustrious industry and entrepreneurial backgrounds. The mode of delivery is a flexible student-centred approach to studying allowing for business and personal life commitments to be considered and planned for in advance of key deadlines. This course shares the institutional Graduate Apprenticeship characteristics; therefore course design: - optimises candidate time within the workplace; - is flexible, enabling students to participate in the course and student experience whilst not bound to a physical campus; - includes meaningful and substantial employer engagement from the outset; - uses innovative teaching, learning and assessment methods; - supports the development of reflective practitioners, equipped to excel in their field and add value to their employers; - ensures the professional practice of learners is informed and enhanced by expert technical and theoretical knowledge; - combines work-based learning, the latest thinking within the field, and work-related/ centric assessments to produce graduates equipped with valuable skills and knowledge. The Graduate Apprenticeship model of achieving a degree through work-based learning has been designed such that:

	- In contrast to the traditional student journey, Graduate Apprenticeship students are employed for the entire duration of the course, and the course is designed to integrate course learning outcomes with suitable workplace projects in collaboration and partnership with their employer. The development of transferable skills is also supported throughout the course via course content, assessment, and work-based learning. - The course utilises the technology investment the University has made in lecture capture and Virtual Learning Environment (VLE) delivery providing the student with a cutting edge experience. Employers are closely involved with the specification and delivery of work-based learning activities to support the learning and development of students. Workplace Mentors appointed by the employer, and supported by the University, work with students to create individual learning plans which outline the shadowing, observation, training, work activities and project opportunities which students undertake in the workplace. These learning plans are reviewed by the University to ensure the required employer contextualisation is appropriate. The course adopts a practical approach to developing students' professional skills; for example employing group discussion activities within course workshops to foster leadership, listening, and communication skills, and requiring students to deliver presentation and reports as a means of further refining their professional skillset. Within the work-based learning environment, students will continue their professional development as they enhance planning and negotiation skills via development of their learning plan in conjunction with their Workplace Mentor and engage in practical activities designed to further their professional skillset. The knowledge and insights gained across the University and work-based learning environment, coupled with the active role students are expected to take in planning and executing their own learning, will act to support the students' overall professional enrichment.
Placement/Professional Experience	Students enrolled on the course are employed on a full or part-time basis (minimum 21 hours per week) by a collaborating partner organisation while studying for the degree. Employers may provide a series of internal placements within their organisations as part of the student's learning plan where it would be beneficial for the student to spend time within a department other than their own to gain specialist skills and knowledge and where this can be supported by their employer.
Opportunities for Further Study	An Honours degree achieved with at least an upper second classification (2.1) will normally provide access to postgraduate and doctoral programmes in UK higher education institutions.
External and Internal Indicators of Quality and Standards	In October 2016 the QAA (Quality Assurance Agency for Higher Education) issued an Interim Statement on Degree Apprenticeships confirming the final UK-wide statement will be in keeping with the existing suite of characteristics statements within Part A of the UK Quality Code for Higher Education. The QAA further state the Quality Code is a key reference point for setting the academic standards and assuring the quality of apprenticeships involving a higher education qualification. Subsequently, in May 2017 the QAA published its paper 'Quality Assuring Higher Education in Apprenticeships: Current Approaches', a second edition of which was published in July 2018. This document highlights considerations for higher education providers in relation to standards and quality and extrapolates relevant guidance from the UK Quality Code. Whilst the document does not constitute formal QAA guidance, its contents have informed the University's approach to the design and delivery of all Graduate Apprenticeships. QAA published a Computing Subject Benchmark Statement in 2016 outlining the standards expected for the award of Honours degrees in computing which, together with the Skills Development Scotland Data Science subject framework, provided the basis for the development of the Graduate Apprenticeship in BSc (Hons) Data Science with Artificial Intelligence. The QAA benchmarking statement for Computing represents the national threshold standards for achievement in this area and articulate the attributes and capabilities, which those possessing qualifications from these areas should demonstrate. As such, QAA benchmarks underpin course articulation, delivery and development through providing guidance for learning outcomes across the breadth of the course and within specific cognate areas such as technology and management. SCQF statements are also used during the design of modules, courses and programmes and the relevant SCQF level is included within each Module Descriptor.

	The University appoints External Examiners to assure standards of assessment. Examiners are appointed from both academic and practice backgrounds to cover key subject areas within Computing. The School has strong links with other academic institutions, both within the UK and Europe, which supports the maintenance of standards at national and international levels. In December 2023, the School had its Chartered IT Professional accreditation renewed by the BCS (British Computer Society) for its undergraduate and postgraduate programmes. The quality assurance procedure includes a regular cycle of accreditation visits by a panel of subject experts appointed by the BCS who evaluate the courses according to strict quality control standards. A number of internal processes and procedures are used by the school to ensure that standards are maintained and enhanced. Stakeholders' views on standards are routinely obtained from the Staff/Student Liaison Committee, Undergraduate Programme Management Team, Student Voice Questionnaires, Industrial Board, industrial visits to placement students, staff involvement in professional body activities and staff participation in accreditation events outside the School and at other HE institutions.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
	Employment: All applicants must be employed in a full or part-time position (minimum of 21 hours per week), relevant to their course of study and have the right to live and work in Scotland. An applicant's employer must be committed to and involved in the provision of a suitable workplace environment, coupled with supportive workplace learning guidance and mentoring to create a suitable workplace environment in which the student will be able to obtain the necessary experience and learning outcomes. Employers must be willing to formally partner with the University via a Collaboration Agreement which sets forth the obligations of each organisation and an individual-level Learning Agreement which details the obligations of the Workplace Mentor; Line Manager if they are not also the Workplace Mentor; Student; and University. Subject specific qualifications: SQA Higher: BBBC and two further passes to include English and Maths at National 5 grade C or above if not held at Higher. Applicants who have successfully completed two Advanced Highers (or equivalent), including Computing and another numerate subject, may be eligible.

Course-Specific Entry Requirements	GCE A Level: BCC and two further passes to include English and Maths at GCSE grade 5/C if not held at A Level. Irish Leaving Certificate Higher: H2H3H3H3 and two further passes to include English and Maths at grade O3 or above if not held at Higher. IB Diploma: 27 to include a pass in Standard Level English and Maths at grade 4 or above if not held at Higher Level. Advanced entry: Applicants who are interested in applying for advanced entry will be considered on an individual basis. Applications may be based upon prior education and relevant work experience. A relevant HNC or HND will be considered for entry into Year 2 or Year 3 with a minimum of 1 to 2 years relevant work experience. Applicants with a relevant Bachelor's degree will be considered for entry to the Honours Stage (Stage 4 of this course) with a minimum of 3 years relevant work experience. Attention is drawn to the University's process for Recognition of Prior Learning. English language requirements: All applicants whose first language is not English are required to have an Academic IELTS (International English Language Testing System) minimum score of 6.0. Appropriate equivalents will be considered.
Alternative Qualifications for Entry	Alternative qualifications: Applications are welcomed from those with equivalent qualifications to those stated, including Foundation Apprenticeships (SCQF 6), SVQs/NVQs, access programmes and overseas qualifications. Applications from students with non- standard qualifications or work experience will be considered on an individual basis.
Interview Procedures	Interviews may be required on a case-by-case basis.
Course-Specific Regulations	Honours classification is based solely on performance in Stage 4 of the course. The main intake for the course is in September. A February intake may exceptionally be available depending on annual funding availability. Please contact the School for further information on intakes.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.

University Strategy	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality. Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The introduction of the Graduate Apprenticeship model is a natural development of the University's philosophy to combine theory, practice and experience in developing the skills, knowledge, confidence and professional practice of its students. The principles of work-based learning are core to the philosophy of each Graduate Apprenticeship course at Robert Gordon University. It is fundamental to the Graduate Apprenticeship courses that students have access to relevant and up-to-date theory as they approach practical workplace situations and problems. Encountering theory and practice simultaneously will enable the student to critically analyse each in conjunction with the other. This will embed a rounded knowledge base and develop the ability to question situations while seeking a range of opinions in forming solutions to challenges. Working in conjunction with learning will also enable students to develop the soft skills necessary in an ever-changing workplace. This means that the student will develop not only the knowledge and skills required for the technical aspects of their career but also the ability to operate in the workplace and communicate effectively with internal and external parties.

	Most of the practical aspects of the course take place in the student's workplace and will therefore vary according to the nature and resources of the student's employing organisation. In addition, the resources of the School are available to the student and support of the teaching staff attached to the course.
	The institutional teaching and learning strategy for Graduate Apprenticeships is characterised by a commitment to ensuring a genuine interaction between theoretical and work-based learning. This is supported by a pedagogical approach which embraces and makes use of experiential learning where experiences framed by reflection lead to learning (Schon, 1983) and andragogy where adult learners understand the need to learn something and self-direct their learning process via a negotiated learning plan. Specifically students are encouraged to consider learning objectives; identify learning opportunities associated with these and reflect upon these experiences to consolidate and apply learning. With regard to work-based learning activities, employers are guided in the identification of suitable activities and projects, and Workplace Mentors are supported in aiding students to reflect on and capture learning from the workplace and encourage the unification of knowledge and experience from the work and university spheres. The Learning Process: Teaching Strategy and Methods The course curriculum is designed to deliver the skills and knowledge identified by Skills Development Scotland's framework for Data Science. It has been developed as a coherent whole with an emphasis on the integration of theory, practical skills and applications. It enables the student to acquire and develop the subject knowledge and understanding necessary for successful professional employment. Close consultation with the student's employer will ensure that practical situations and projects encountered within the workplace correlate with the theoretical knowledge developed throughout the course. Opportunities for critical reflection and analysis of apparent or real disparities or conflicts between theory and practice will inform the student of the practicalities of real-world situations and form a rich basis for developing their professional practice. These skills are developed at levels corresponding to the Scottish Credit and Qualifications Framework levels and form a progression through levels 7, 8, 9 and 10. As the student progresses through the course they are encouraged to expand their understanding and critical appreciation of all aspects relating to the study of Data Science with AI. A central feature of all Robert Gordon University Graduate Apprenticeship courses is that they promote a unified, integrated view of the subject area. Courses are designed to: - Present a coherent underpinning of the theory and practice appropriate to the aims of the course which is further developed and applied within the work-based learning environment. This enables graduates of the course to adapt to future developments in the field. - Reflect changes within the field and ensure that coverage is given to a selection of emerging topics so that students are aware of likely future developments in the subject together with their potential impact. - Integrate the delivery of teaching across thematic boundaries to highlight fundamental aspects of interconnectivity and interdisciplinarity within different areas of the syllabus. - Integrate modes of assessment which complement the methodology of work-based learning and support the development of workplace communication skills. - Facilitate students to exhibit competence and achievement across a number of different themes (integrated assessment). - Encourage students to combine elements of their learning from different parts of the course and to show their accumulated and integrated knowledge and understanding of a topic or subject area (synoptic assessment). The professionally-orientated nature of the course requires students to obtain general and specific skills, this being addressed by a combination of teaching and learning methods. These methods are built around mentored workplace learning directed, supported and extended by University guided, student-centred learning including virtual lectures, tutorials, discussion forums, workshops and private study.

Teaching and Learning Strategy

Teaching and Learning Approaches

The main goal of the teaching and learning strategy for the University's Graduate Apprenticeships is to enable students to become critical, reflective, independent learners who are active participants in the learning process and who develop the motivation and autonomy needed to take control and regulate their own educational experiences and develop the discipline of continuous professional development and life-long learning. This means that, throughout their course of study, students will begin to take more responsibility for their own acquisition of necessary skills and knowledge so that, by the end of their degree, they will be in a position to evaluate and assess their own effectiveness.

This process is driven by various objectives such as the need for constructive alignment of syllabus and learning approaches and makes use of a wide-range of appropriate delivery styles. In addition to mentored workplace learning these include conventional instructional teaching methodologies, social learning activities such as group discussion, as well as constructivist, investigative approaches such as problem-based learning and enquiry-based learning.

The delivery of material centres on using the University's VLE, CampusMoodle, virtual campus platform. This centres on the creation and sustenance of active community groups, and delivery of topics is structured in such a way as to facilitate the maintenance of student interaction. Papers, supported by recommended reading lists and private study guides, are augmented by a discussion forum and the initiation of discussion threads. Live chat rooms are used to encourage debate and the exchange of ideas between students, on the one hand, and with tutors on the other. Video conferencing is used, as required, during specific workshops and presentation/discussion of project work. The Library provides support for all students and is progressively expanding its provision of electronic texts such as e-journals and e-books.

Modes of Delivery

Work-Based Learning

The workplace forms the central learning environment for the student. It is important, however, to appreciate that whilst much learning takes place naturally in a workplace that this typically happens in an unplanned and unstructured way, with limited opportunity for reflection or consolidation of learning. For the purposes of this course there needs to be a structure in place to guide the student and ensure that learning outcomes are met. This structure is governed by a subject framework established by Skills Development Scotland and developed into a curriculum by the University who work closely with the student's employer to ensure that learning outcomes are achieved through correlated theory and practice. This is expressed for the student in the form of a learning plan which guides them through the course and helps to ensure that workplace learning is supported, resourced, and planned with space and time for critical reflection and consolidation of learning.

Workplace Mentoring

Key to the delivery and support of work-based learning is the Workplace Mentor. Partner employers are required to assign a Workplace Mentor whose role is to oversee the work-based learning undertaken by the student. The University provides employers with guidance as to how appropriate Workplace Mentors should be selected, and provide Workplace Mentors with training and support via induction and ongoing liaison designed to help mentors understand the remit of their role and the activities associated with this. Workplace Mentors have an overview of the course and its aims, develop the student's learning plan, assign projects and in-house placements and provide feedback on work-based activities. Workplace Mentors are granted 'Affiliate Student' status enabling them to access all course learning materials via the University's VLE. In addition to mentoring support for students, supplementary support may be provided by the employer, for example when subject matter experts are brought in to provide specific business or technical expertise or through access to other departments within the organisation. This enables work-based learning to reflect the range of the curriculum as the student progresses in their studies, therefore providing adequate support for the student's workplace learning on a project, placement or task basis.

Employers' Responsibilities

Partner employers commit to supporting students via the signing of a Collaboration Agreement which sets forth the obligations of the employer and the University. In supporting students, employers are obliged to provide a work-based learning environment which adequately represents and reflects the students' course to ensure appropriate functional expertise, supporting infrastructure and professional and technical knowledge within the organisation. Central to this is the provision of a supportive and educative work-based learning environment which affords opportunities to learn, gain new experiences, acquire and build competencies, and reflect on and apply learning. Additionally, employers are obliged to provide protected time to allow students to attend on-campus study days and to undertake study-related activities such as participation in online discussion forums, and access to the requisite IT hardware and software to facilitate this. Workplace Mentors will also be requested to consent to and sign a Learning Agreement alongside the University and the student, which will outline the mutual responsibilities and commitments of the key parties involved in achieving the aim of the student obtaining a degree via work-based learning and University-delivered course content at an individual level. In the event that the Mentor is not also the Line Manager of the student, the Line Manager will also be requested to agree to and sign the Learning Agreement.

Student Role in Learning

Students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research and self-directed enquiry. As students develop over the duration of the course, they are expected to become increasingly self-directed and self-motivated, taking greater control of the study process, and defining their own personal objectives for learning. Students are encouraged to reflect regularly upon their learning and to establish a view of their own progress and development, which they triangulate with feedback from their Workplace Mentor and course delivery team. Students are expected to utilise the full range of facilities and resources available to them, for example the services and resources provided by the library and the support available via the Graduate Apprenticeship Success Coach.

Lectures

Lectures are for the purpose of dissemination of information and the demonstration of techniques. These are usually be available via the VLE but may also be delivered face to face at the University as part of an on-campus event.

Tutorials

Tutorials may be arranged remotely (e.g. online via the VLE) or in person. Tutorials are an opportunity to raise questions, discuss ideas and consequences and exchange experiences and lessons learned. This format is used for a wide range of activities, each suited to the particular subject. It may consist of the staff supporting students engaged in problem-solving or may consist of a series of group exercises where each group is encouraged to allocate responsibilities, allocate tasks, etc. Generally this type of teaching is used to support topic content, clarify the material and experiment with the techniques and skills required.

On-Campus Events

Beyond the initial course induction, Graduate Apprenticeship students are expected to attend 1 to 2 on-campus day per module of study. The format of these days vary accordingly to reflect course content and student needs, but generally these may be group sessions where students can discuss the relationship between theory and practice through critical analysis and reflection. On-campus events also give students the opportunity to develop their professional practice through ideas exchange with peers and academic staff and through other networking. On-campus events are scheduled within a course calendar which is publicised to employers and students in advance of course commencement.

Individual Project

The student will agree a specification for this extended piece of work with their University tutor and their Workplace Mentor to fulfil the requirements of the Honours Individual Project. The academic work will be completed under the guidance and supervision of the student's assigned University tutor.

Assessment	A variety of assessment methods are used in this course, as deemed appropriate to the learning outcomes and subject material being assessed and the context in which the material is delivered. Assessment will be in line with the University Assessment Policy and quality assurance and will remain the responsibility of the University. Coursework Coursework may be used as a form of summative assessment, and within the context of Graduate Apprenticeship courses this may be work-related, for example based around a project being undertaken in the workplace, or work-centric, such as a report in response to a case study scenario where the case study is built to reflect current issues within a specific sector. Coursework may take many forms, for example reports, essays, posters to showcase student proposals, reflective statements. Coursework may also include portfolios of evidence demonstrating the application of specific skills and knowledge allowing a view to be formed of the extent to which a student has satisfied learning outcomes. The University provides students with access to ePortfolio software in order to assist students in their creation of portfolios. Where a deadline applies to a particular piece of work the student will be informed at the commencement of the relevant module. Appropriate review measures will be implemented in order to ensure that the students' coursework is individual in nature and not the result of a group effort involving colleagues. Work-Based Learning Assessment Amongst the assessment tools suitable for the context of work-based learning are diary logs, student blogs, vlogs, other social media engagement, logs developed through critical analysis and reflection which can be worked up as case studies, written critical analysis of theory and practice, reports and presentations. As discussed above, portfolio work also provides an invaluable self-compiled reference resource and starting point for additional learning for the student as they progress in their professional practice. Whilst the output of work-based learning may form the basis for assessment, Workplace Mentors are not expected to play a role in the evaluation of work submitted to Robert Gordon University for summative assessment purposes. Computer Aided Assessment Computer-assisted assessment packages integrated into the VLE may be used for formative and/or summative assessment, to enhance student engagement and increase the speed of feedback. Unseen Exams Unseen exams may be used to assess students' base of knowledge and understanding. Individual Project Projects are used in later stages of study as a learning vehicle and assessment process which allows students to tackle more complex problems that often require the integration of knowledge and understanding, and intellectual skills from a variety of subject areas. As part of achieving an honours classification students will undertake an Honours Individual Project which is an independent piece of research relating to a workplace topic relevant to the subject of Data Science. The findings from the project should shape change in the student's place of employment enhancing the project/team/departmental/business operation. The individual project will be assessed through a report, presentation and a demo. Assessment Types
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	Many modules use a combination of assessment types that integrate modes of assessment in order to balance the types of activity and purposes of assessment, such as formative and summative. Where there are a number of modules that cover allied subject material, integrated assessment may be used to facilitate students to exhibit competence and achievement across a number of different themes. At all Stages, synoptic assessments embedded into projects and other practical assignments are used to encourage students to combine elements of their learning from different parts of the course and to show their accumulated knowledge and understanding of a set of topics or a subject area. A student's Honours Classification is based solely on performance in Stage 4 of the course. The Assessment of Student Work The University has developed a Grading Scheme for assessment purposes, details of which may be accessed at: http://www.rgu.ac.uk/academicregulations . It is important to note that all grades indicated to students during the course of the academic session are provisional until confirmed by an Assessment Board. The Board consists of academic staff involved in course delivery as well as External Examiners, and a final decision is confirmed on the performance of every student for each module undertaken by them and any eligibility for an award. This is subsequently confirmed to each student in a published transcript, and may be accessed electronically on the Student Portal. Any potential barriers to students with disabilities or specific needs are assessed by the University Inclusion Centre who liaise with the School and/or appropriate teaching staff to ensure that such students are not disadvantaged during an assessment. In the case of examinations, students must register by a specified deadline date to have specific examination requirements considered.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • an ongoing scheme of personal/pastoral support for students, including online support, provided by the <i>Inclusion Centre</i>; • an extensive programme of student study skills delivered through <i>Student Life</i> and the <i>Library</i>; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in the Quality Assurance Agency's Quality Enhancement Themes, enhancementthemes.ac.uk ; • support for staff from the <i>Department of Academic Quality and Learning Enhancement (AQLE)</i>, including credit rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub;

	• Moodle, http://campusmoodle.rgu.ac.uk/ , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
Course-Specific Support for Teaching, Learning and Assessment	The course will have a dedicated Course Leader to support the development and the delivery of the course. Module Coordinators and tutors are also available to support students with course and module related issues and concerns. Dedicated administrative support will be available to deal with any queries students may have. The School has dedicated flexible learning labs which will be available to students during relevant periods in their studies. The School's e-learning staff will also support students and staff as required. Students undertaking individual projects are provided with a detailed Project Handbook covering all aspects of the project operation and assessment. Extensive written guidance is also available via the VLE. All students have an account on the University computer network and can access PCs in the school and across the University. Specialist software is provided on PCs and through virtual machines in the School. The support staff also assist students with administrative queries. Frequent informal feedback is provided to students through regular contact with academic staff and Workplace Mentors. Feedback is provided on all assessed work normally within 20 working days of submission of the work. Provisional grades are made available electronically with final grades available electronically via the Student Portal following ratification by the Assessment Board.
University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.

Course-Specific Monitoring of Quality and Standards	Annual Course Appraisals are reviewed and approved by the Course Management Team and the School Academic Board which meet at least once per semester. The Course Management Team considers teaching and learning issues arising from feedback generated by External Examiner Annual Reports, staff/student engagement/partnership liaison, and Student Voice Questionnaires.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ; • staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	The Staff Student Liaison Committee meets at least once per semester. The membership of the Committee consists of the Course Leaders, administrators, Student School Officers and student representatives from each Stage of the undergraduate courses. The committee meeting is chaired by the Student School Officer - who also is a member of the School Academic Board - and student representatives are asked to collect comments from their class. However, student representatives are encouraged to liaise with Course Leaders/Module tutors when issues arise (rather than waiting for the next Staff Student Liaison Committee meeting). Minutes of the meetings are posted on Moodle for all students on the course to access.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Higher Apprenticeship in Certificate of Higher Education - Higher Apprenticeship in CertHE	Data Science	Aberdeen, United Kingdom	Exit Award		120

Stage 2

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Higher Apprenticeship in Diploma of Higher Education - Higher Apprenticeship in DipHE	Data Science	Aberdeen, United Kingdom	Exit Award		240

Stage 3

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Graduate Apprenticeship in Bachelor of Science - Graduate Apprenticeship in BSc	Data Science with Artificial Intelligence	Aberdeen, United Kingdom	Exit Award		360

Stage 4

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Graduate Apprenticeship in Bachelor of Science with Honours - Graduate Apprenticeship in BSc (Hons)	Data Science with Artificial Intelligence	Aberdeen, United Kingdom	Target Award		480

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	The organisation's business environment and its data strategy, including an understanding of how structured and unstructured data is sourced, collected and integrated, and how business requirements drive the creation of data architecture.	SC1	
LLO2	The range of machine learning and statistical techniques used in data analytics to help businesses gain insight from data including trends, patterns and forecasting, and the communication of those insights through visualisations appropriate to the intended audience.	SC1	
LLO3	The key principles and skills required to develop intelligent systems to drive business efficiency and support decision-making through real-world applications such as recommender systems, virtual assistants, web and social media mining and image analysis.	SC1	
LLO4	The role of big data systems in meeting the business requirements for scalability and high performance and availability to deal with an ever increasing need to store and process high volumes of data in real or near real time.	SC1	
LLO5	Key legislation and ethics in data science, and how they can affect the way data is collected, stored and processed in compliance with data security standards and legislation including organisational standards and policies.	SC1	
LLO6	Apply mathematical and statistical skills to model and analyse a range of problems relevant to data science, and develop transferable analytical and problem skills of value to the workplace.	SC2	
LLO7	Demonstrate an applied knowledge of specialised skills, techniques, practices and tools in the implementation of business intelligence and data visualisation solutions.	SC2	
LLO8	Apply knowledge and skills required to integrate structured and unstructured data from various heterogeneous sources into a single source of truth to provide business insights and support decision-making.	SC2	
LLO9	Demonstrate an applied knowledge of specialised skills in the development of intelligent systems using relevant machine learning techniques to solve real-world problems and an awareness of strengths and limitations of such technology.	SC2	
LLO10	Apply knowledge, skills and understanding of specialist areas in data science, including intelligent information retrieval, natural language processing and text and image analysis and their use in real-world applications, such as web and social media mining.	SC2	
LLO11	Demonstrate practical skills in the construction of databases and big data solutions, including the appraisal of different models and platforms to meet business requirements.	SC2	
LLO12	Problem-solve and critically analyse facts and requirements to determine the cause of a problem and identifying and selecting appropriate solutions.	SC3	
LLO13	Research, analyse and evaluate a range of business data, sources of information and appropriate methodologies, and to use that research for evidence-based decision-making.	SC3	
LLO14	Critically evaluate and test the extent to which a data science solution meets the business requirements and criteria defined for its current use and future development.	SC3	
LLO15	Innovate, create and be enterprising, including the ability to act entrepreneurially to generate, develop and communicate ideas, manage and exploit intellectual property, gain support, and deliver successful outcomes.	SC3	
LLO16	Network, demonstrating an awareness of the interpersonal skills of effective listening, negotiating, persuasion and presentation.	SC3	
LLO17	Foster reflective thinking and critical judgement to logically analyse issues, evaluate different options and implement decisions.	SC3	

LLO18	Recognise the professional, economic, social, environmental and ethical issues involved in the sustainable exploitation of data science technology and be guided by the appropriate professional, ethical and legal practices.	SC3	
LLO19	Develop numeracy skills, including the use of quantitative skills to manipulate data, evaluate, estimate and model computing problems and phenomena.	SC4	
LLO20	Work collaboratively both internally and with external stakeholders to develop awareness of mutual interdependence, team building, leadership and motivation of others.	SC4	
LLO21	Develop skills in the design, implementation, and deployment of data science systems and evaluate their impact on organisations.	SC4	
LLO22	Develop skills in communicating data science results to a range of audiences.	SC4	
LLO23	Demonstrate the ability to take responsibility for own learning and to exhibit reflection on individual actions, or within group situations in a collaborative and professional manner.	SC5	
LLO24	Build and maintain relationships during their career.	SC5	
LLO25	Develop emotional intelligence and empathy.	SC5	
LLO26	Develop self-analysis and an awareness to diversity in people and cultures.	SC5	
LLO27	Plan, manage and communicate the outcomes of a data science project.	SC5	
LLO28	Demonstrate familiarity with the basic techniques used throughout the data analytics lifecycle (data cleaning, transformation, analysis and visualisation) to solve business problems.	SC1	SCQF 7
LLO29	Demonstrate some knowledge and understanding of the mathematical principles and techniques behind data science, which will form the basis of key analytical and problem solving transferable skills.	SC1	SCQF 7
LLO30	Demonstrate a working understanding of the fundamental principles and techniques of programming including aspects of the software development lifecycle (design, implementation and testing) to solve business problems.	SC1	SCQF 7
LLO31	Demonstrate an understanding of the roles and responsibilities for data science in the organisation and demonstrate effective work habits, including time management and prioritisation.	SC1	SCQF 7
LLO32	Study the subject in a systematic, directed way with the aid of appropriate academic and professional supervisory guidance within classroom, laboratory and working environments.	SC2	SCQF 7
LLO33	Apply analytical and modelling skills to a range of simple problems relevant to data science.	SC2	SCQF 7
LLO34	Apply statistical techniques to a variety of data sets relevant to the business context.	SC2	SCQF 7
LLO35	Apply industry standard tools to manage small software deliverables.	SC2	SCQF 7
LLO36	Use and apply some of the basic professional skills, techniques, and practices associated with computing in a business environment, including how to meet personal and organisational objectives.	SC2	SCQF 7
LLO37	Apply the basic data science principles and procedures to standard, simple situations.	SC3	SCQF 7
LLO38	Demonstrate the ability to analyse situations and problems, based on the set of classifications presented by tutors and professional colleagues.	SC3	SCQF 7
LLO39	Systematically relate to a limited number of facts/ideas/elements in an imitative manner.	SC3	SCQF 7
LLO40	Start to form their own value judgements based on criteria provided.	SC3	SCQF 7
LLO41	Apply the basic tools/methods of data science to simple, standard problems.	SC3	SCQF 7

LLO42	Demonstrate aspects of data science in a clear and concise way, in writing and orally, in relatively limited-length pieces of work, and be able to communicate the basic design and documentation of programs appropriately.	SC4	SCQF 7
LLO43	Demonstrate essential numeracy and mathematical techniques and the ability to manipulate data at a level appropriate to this stage of the degree, and develop an understanding of basic concepts in logic and computation.	SC4	SCQF 7
LLO44	Demonstrate practical skills in computer science using basic ICT applications as well as those skills necessary for efficient use in a business environment.	SC4	SCQF 7
LLO45	Learn in an effective and purposeful way, showing the beginnings of development as an autonomous learner, and exercise some initiative and independence in carrying out defined activities.	SC5	SCQF 7
LLO46	Interact effectively with colleagues, mentors, managers, tutors and fellow students via data science development tasks, problem-solving tutorials and the production of appropriate documentation.	SC5	SCQF 7
LLO47	Work, under guidance, with others to acquire an understanding of current professional practice.	SC5	SCQF 7
LLO48	Demonstrate an understanding of the importance of learning strategies and techniques in their own development and life-long learning, and for corporate learning and development, and understand the principles of personal development planning by creating, implementing and maintaining a personal development portfolio and a personal action plan.	SC5	SCQF 7
LLO49	Demonstrate an understanding of database design principles and their application to meet business requirements, including data governance and security as well as the legal issues around data privacy.	SC1	SCQF 8
LLO50	Understand the main principles underlying data mining, including knowledge of the strengths and limitations of this technology and an awareness of how data mining may exhibit biases and prejudice.	SC1	SCQF 8
LLO51	Demonstrate an understanding of advanced programming techniques used in software development, and an appreciation of the properties of good software design.	SC1	SCQF 8
LLO52	Demonstrate an understanding of the application of ethical theories in the IT business environment, and demonstrate a high level of professional and ethical conduct.	SC1	SCQF 8
LLO53	Apply current thinking regarding ethical practices within a data science and computing context in order to appraise a variety of approaches within industry.	SC2	SCQF 8
LLO54	Apply principles of good design to software development in line with business requirements and industry standards.	SC2	SCQF 8
LLO55	Create a relational database schema that incorporates key integrity and business constraints, while ensuring data security and privacy.	SC2	SCQF 8
LLO56	Apply and adapt appropriate data mining techniques to a given problem, including the selection of an appropriate evaluation technique.	SC2	SCQF 8
LLO57	Apply standard principles and procedures to somewhat more demanding situations.	SC3	SCQF 8
LLO58	Demonstrate increasing ability to apply classifications and analyse relatively simple situations.	SC3	SCQF 8
LLO59	Demonstrate an appreciation of the need for relating and organising a range of facts/ideas/elements in an argued case.	SC3	SCQF 8
LLO60	Identify a range of valid alternative solutions, and begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way.	SC3	SCQF 8
LLO61	Apply given tools/methods accurately and carefully to more demanding problems appropriate to this stage.	SC3	SCQF 8
LLO62	Develop communication skills in styles appropriate to the setting, e.g. professional or formal academic, and in more extended pieces of work.	SC4	SCQF 8
	Apply a selection of more advanced numerical, modelling and design skills, relevant to specialised areas of data mining, software development,		

LLO63	and the design and management of database systems.	SC4	SCQF 8
LLO64	Demonstrate, use and access a broad selection of more advanced data science and IT skills in a wide range of contexts in all topics.	SC4	SCQF 8
LLO65	Exercise autonomy and initiative in some activities at a professional level within the range of exercises set by tutors and supervisors and other professional colleagues.	SC5	SCQF 8
LLO66	Take the lead on planning work activities in familiar or defined contexts.	SC5	SCQF 8
LLO67	Practise in ways that show awareness of own and others' roles, responsibilities and contributions when carrying out and evaluating tasks.	SC5	SCQF 8
LLO68	Work, under guidance, with others to acquire an understanding of current legal, ethical and professional issues, and manage these issues in accordance with current professional and/or ethical codes of practice.	SC5	SCQF 8
LLO69	Apply analytical skills to technology solutions development and to systematically analyse and apply structured problem solving techniques to them.	SC5	SCQF 8
LLO70	Demonstrate an understanding of the state-of-the-art big data analytics concepts and techniques, including modern parallel computation methodologies for designing, implementing and evaluating scalable solutions.	SC1	SCQF 9
LLO71	Demonstrate an understanding of the key techniques required for the development of intelligent systems which are capable of learning to solve real-world problems.	SC1	SCQF 9
LLO72	Demonstrate a critical understanding of the strengths and limitations of artificial intelligence and machine learning technology and an awareness of related ethical issues.	SC1	SCQF 9
LLO73	Demonstrate an understanding of business intelligence concepts, methods and techniques for solving business problems, including data warehousing and data integration methodologies.	SC1	SCQF 9
LLO74	Understand the nature of business innovation and the skills required for developing creative business ideas.	SC1	SCQF 9
LLO75	Apply knowledge, skills and understanding in the analysis, design, implementation and evaluation of big data systems, including NoSQL data stores and parallel computation frameworks.	SC2	SCQF 9
LLO76	Demonstrate an applied knowledge of some specialised or advanced skills, techniques, practices and tools in the implementation of business intelligence solutions.	SC2	SCQF 9
LLO77	Apply data visualisation techniques appropriate to the intended audience, while showing an awareness of accessibility issues.	SC2	SCQF 9
LLO78	Apply knowledge, skills and understanding of data integration techniques, including the extraction, transformation and loading of large volumes of data from various sources into a data warehouse in order to provide insights for the business.	SC2	SCQF 9
LLO79	Demonstrate an applied knowledge of specialised skills in the development of intelligent systems using relevant machine learning techniques to solve real-world problems and an awareness of strengths and limitations of such technology.	SC2	SCQF 9
LLO80	Design and undertake appropriate market, competition and customer research in order to evaluate the feasibility of a proposed business initiative.	SC2	SCQF 9
LLO81	Practise routine methods of enquiry and research.	SC2	SCQF 9
LLO82	Demonstrate the ability to select and use principles and procedures appropriate to the situation or problem in hand.	SC3	SCQF 9
LLO83	Recognise familiar ideas or principles in new contexts or situations and analyse them systematically and effectively.	SC3	SCQF 9
LLO84	Show an ability to relate and organise facts, ideas and elements in an argued case and relate these ideas to a wider range of situations.	SC3	SCQF 9

LLO85	Identify a range of valid alternative solutions, begin to discriminate and evaluate in a reasoned, systematic and increasingly independent way.	SC3	SCQF 9
LLO86	Begin to self-direct, identify key elements of problems, and choose appropriate methods to construct their solution, with only general guidance, in all topics.	SC3	SCQF 9
LLO87	Engage effectively in a variety of roles, to produce clear, well-structured reports and presentations on relevant aspects of data science systems.	SC4	SCQF 9
LLO88	Apply a selection of specialist numerical, modelling and design skills, relevant to the design and implementation of intelligent and big data systems.	SC4	SCQF 9
LLO89	Apply a selection of specialist business intelligence tools to support decision-making.	SC4	SCQF 9
LLO90	Access and successfully use a broad selection of specialist computing and ICT skills in a wide range of contexts, in all topics.	SC4	SCQF 9
LLO91	Adopt a broad-ranging and flexible approach to study, identifying learning needs and pursuing activities designed to meet these needs in increasingly autonomous ways.	SC5	SCQF 9
LLO92	Assume an appropriate role in the design, implementation, evaluation, and management of data science projects.	SC5	SCQF 9
LLO93	Use negotiation and collaboration skills to manage the interface between data science and other functional areas of an organisation, and the users of its computer systems.	SC5	SCQF 9
LLO94	Deal with legal, ethical and professional issues in accordance with current codes of best practice, seeking guidance where appropriate.	SC5	SCQF 9
LLO95	Be able to put forward, demonstrate value and gain commitment to a moderately complex technology-oriented solution, demonstrating understanding of business needs, using open questions and summarising skills and basic negotiating skills.	SC5	SCQF 9
LLO96	Demonstrate an understanding of the relationship between the principal specialist areas of the course and integrate them in the Honours Individual Project as a culmination of the degree programme.	SC1	SCQF 10
LLO97	Demonstrate appropriate IT project management expertise, professional and ethical conduct, and in-depth technical, problem solving and communication skills in the conduct of the Honours Individual Project and in-line with the expectations of the workplace.	SC1	SCQF 10
LLO98	Demonstrate confident familiarity and critical understanding of the defining concepts and features of intelligent technologies for information retrieval and natural language processing.	SC1	SCQF 10
LLO99	Demonstrate an understanding of the key concepts of computer vision and the skills required to analysing images and video.	SC1	SCQF 10
LLO100	Demonstrate a commitment to personal development in data science, through a critical review of the latest developments in the discipline (techniques, platforms, software tools, etc.).	SC1	SCQF 10
LLO101	Apply a practical knowledge of data science to execute a defined and substantial computing project involving research and development within the context of the work-based learning environment for the Honours Individual Project.	SC2	SCQF 10
LLO102	Be able to conduct an effective, in-depth critical review of the literature to inform the development of a solution to the problem tackled as part of the Honours Individual Project.	SC2	SCQF 10
LLO103	Apply knowledge, skills and understanding of specialist areas in data science, including intelligent information retrieval, natural language processing and text analytics and their use in real-world applications, such as web and social media mining.	SC2	SCQF 10
LLO104	Demonstrate practical skills in the construction of computer vision solutions, using machine learning and deep learning methods.	SC2	SCQF 10
LLO105	Be able to keep up to date with the latest trends and applications in data science (e.g., virtual assistants, sentiment analysis, recommender systems and mining of real-time data), and show awareness of the challenges to data science posed by real-world applications.	SC2	SCQF 10
	Carry out confident and accurate selection and application of principles and procedures to the solution of a range of professional situations and		

LLO106	problems, working with increasing autonomy in the work-based environment.	SC3	SCQF 10
LLO107	Identify and classify principles and ideas in new contexts and situations, and analyse them systematically, effectively and critically based on work-based experience gained.	SC3	SCQF 10
LLO108	Bring together facts, ideas and elements in support of a well-structured argument, design novel solutions and evolve new concepts, working with increasing autonomy in the work-based environment.	SC3	SCQF 10
LLO109	Critically evaluate theory, process, solutions and outcomes based on experiences within the work-based learning environment.	SC3	SCQF 10
LLO110	Identify and define complex problems in an increasingly independent, confident and flexible way and apply knowledge and skills appropriate to their solutions based on work-based learning experiences, and at the threshold of professional competence in all topics.	SC3	SCQF 10
LLO111	Demonstrate use and access a broad selection of specialist computing skills, in all topics, but particularly within the Honours Individual Project.	SC4	SCQF 10
LLO112	Make effective use of specialist computing and data science methodologies, and develop appropriate skills for this purpose within the work-based environment.	SC4	SCQF 10
LLO113	Use effective means of communication to express complex ideas to colleagues through appropriate media, based on requirements of the work-based learning context.	SC4	SCQF 10
LLO114	Engage effectively in independent roles, debate in a professional manner, and produce detailed critiques and coherent project reports underpinned by appropriate metrics.	SC4	SCQF 10
LLO115	Give confident oral and formal presentations to informed audiences in a wide range of contexts, particularly within the Honours Individual Project.	SC4	SCQF 10
LLO116	Demonstrate skills, knowledge and understanding through investigative, modelling and problem-solving activities, and manage the process relating to the development, maintenance and evaluation of data science systems appropriate to the scaffolded sense of autonomy within the work-based learning environment.	SC5	SCQF 10
LLO117	Maintain monitoring of standards and quality control as applied to the specification, production, implementation and maintenance of data science systems in accordance with current professional and ethical codes and practices.	SC5	SCQF 10
LLO118	Manage own learning with minimal guidance, using a wide range of resources appropriate to the profession as a whole and specifically within the work-based learning environment, while seeking and making effective use of feedback provided both from the work-based learning environment and academically; at this level within all topics, but particularly within the Honours Individual Project.	SC5	SCQF 10
LLO119	Interact effectively within groups in all topics, and, where appropriate, develop interaction and negotiating skills with customers and other stakeholders.	SC5	SCQF 10

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
		SCQF 7 - SCQF 7

SC2	Practice: Applied knowledge, skills and understanding	SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Full Time Work Based Learning	

Stage 1 / Semester 1**Full Time Work Based Learning**

<i>Core</i>								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM1705	Fundamentals of Programming [Approved]		Lee Christie (lc8168)	Yes	2.0	30	Level 7
Full Time Work Based Learning	CM1706	Introduction to Business Analytics [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 7

Stage 1 / Semester 2**Full Time Work Based Learning**

<i>Core</i>								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM1704	Analytical and Problem Solving Skills [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 7
Full Time Work Based Learning	CM1707	Professional Development [Approved]		Tiffany Young (ty7997)	Yes	2.0	30	Level 7

Stage 2 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM2710	Advanced Programming [Approved]		Mark Bartlett (mb9400)	Yes	2.0	30	Level 8
Full Time Work Based Learning	CM2709	Databases and Information Security [Approved]		Hatem Ahriz (cmsa)	Yes	2.0	30	Level 8

Stage 2 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM2705	Business Ethics and Management [Approved]		Niccolo Capanni	Yes	2.0	30	Level 8
Full Time Work Based Learning	CM2712	Data Mining [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 8

Stage 3 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM3708	Big Data Systems [Approved]		Kit-ying Hui (cmskhu)	Yes	2.0	30	Level 9
Full Time Work Based Learning	CM3710	Machine Learning and Artificial Intelligence [Approved]		Hatem Ahriz (cmssha)	Yes	2.0	30	Level 9

Stage 3 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM3704	Business Ideas and Opportunities [Approved]		Chioma Onoshakpor (co11164)	Yes	2.0	30	Level 9
Full Time Work Based Learning	CM3709	Business Intelligence [Approved]		Jean-Claude Golovine (jrrg5708)	Yes	2.0	30	Level 9

Stage 4 / Semester 1**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM4706	Advanced Data Science [Approved]		Hatem Ahriz (cmsa)	Yes	2.0	30	Level 10
Full Time Work Based Learning	CM4708	Data Science Trends and Applications [Approved]		Hatem Ahriz (cmsa)	Yes	2.0	30	Level 10
Full Time Work Based Learning	CM4700	Honours Individual Project [Approved]	Runs over both semesters	Mark Zarb (mz7371)	Yes	2.0	30	Level 10

Stage 4 / Semester 2**Full Time Work Based Learning**

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Full Time Work Based Learning	CM4709	Computer Vision [Approved]		Kit-ying Hui (cmskhu)	Yes	2.0	30	Level 10

Course Deliveries

Stage	Semester	Delivery Type	Delivery Range	Code	Title	Notes	Owner	Allow Mapping	Version	Credits	Module Level
1	Semester 1	Core	Full Time Work Based Learning	CM1705	Fundamentals of Programming [Approved]		Lee Christie (lc8168)	Yes	2.0	30	Level 7
1	Semester 1	Core	Full Time Work Based Learning	CM1706	Introduction to Business Analytics [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 7
1	Semester 2	Core	Full Time Work Based Learning	CM1704	Analytical and Problem Solving Skills [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 7
1	Semester 2	Core	Full Time Work Based Learning	CM1707	Professional Development [Approved]		Tiffany Young (ty7997)	Yes	2.0	30	Level 7
2	Semester 1	Core	Full Time Work Based Learning	CM2710	Advanced Programming [Approved]		Mark Bartlett (mb9400)	Yes	2.0	30	Level 8
2	Semester 1	Core	Full Time Work Based Learning	CM2709	Databases and Information Security [Approved]		Hatem Ahriz (cmsha)	Yes	2.0	30	Level 8
2	Semester 2	Core	Full Time Work Based Learning	CM2705	Business Ethics and Management [Approved]		Niccolo Capanni	Yes	2.0	30	Level 8
			Full								

2	Semester 2	Core	Time Work Based Learning	CM2712	Data Mining [Approved]		David Lonie (dpl0795)	Yes	2.0	30	Level 8
3	Semester 1	Core	Full Time Work Based Learning	CM3708	Big Data Systems [Approved]		Kit-ying Hui (cmskhu)	Yes	2.0	30	Level 9
3	Semester 1	Core	Full Time Work Based Learning	CM3710	Machine Learning and Artificial Intelligence [Approved]		Hatem Ahriz (cmsha)	Yes	2.0	30	Level 9
3	Semester 2	Core	Full Time Work Based Learning	CM3704	Business Ideas and Opportunities [Approved]		Chioma Onoshakpor (co11164)	Yes	2.0	30	Level 9
3	Semester 2	Core	Full Time Work Based Learning	CM3709	Business Intelligence [Approved]		Jean-Claude Golovine (jrrg5708)	Yes	2.0	30	Level 9
4	Semester 1	Core	Full Time Work Based Learning	CM4706	Advanced Data Science [Approved]		Hatem Ahriz (cmsha)	Yes	2.0	30	Level 10
4	Semester 1	Core	Full Time Work Based Learning	CM4708	Data Science Trends and Applications [Approved]		Hatem Ahriz (cmsha)	Yes	2.0	30	Level 10
4	Semester 1	Core	Full Time Work Based Learning	CM4700	Honours Individual Project [Approved]	Runs over both semesters	Mark Zarb (mz7371)	Yes	2.0	30	Level 10

4	Semester 2	Core	Full Time Work Based Learning	CM4709	Computer Vision [Approved]		Kit-ying Hui (cmskhu)	Yes	2.0	30	Level 10
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Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
Aberdeen, United Kingdom	Work-Based	Full Time	

Sources of Further Information

#	URL and Description
1	https://www.apprenticeships.scot/ Further information about Degree Apprenticeships in Scotland can be found at the Skills Development Scotland Apprenticeships website: https://www.Apprenticeships.scot/ RGU Graduate Apprenticeship information made available to employers and prospective students can be found at: http://www.rgu.ac.uk/business-services/graduate-Apprenticeships Further information about the undergraduate programme in the School of Computing, Engineering and Technology can be found at the School website: http://www.rgu.ac.uk/areas-of-study/subjects/computing/study-options/undergraduate Library resources can be accessed via the Library web portal: https://www.rgu.ac.uk/staff-and-current-students/library Advice for students about study and welfare issues is found at the Student Advice and Support portal: https://www.rgu.ac.uk/student-life/student-advice-and-support/ student-advice-and-support Links to the Academic Regulations governing the course can be accessed at: http://www.rgu.ac.uk/academicregulations